SUPERSEDING
MIL-G-13494B(WC)
31 May 1968

MILITARY SPECIFICATION

GUN, FLUID, DIRECT DELIVERY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a fluid gun (hereafter referred to as gun) used in the delivery of hydraulic fluid to hydro-pneumatic recoil mechanisms and equilibrators.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

UU-T-106	- Tape, Pressure-Sensitive Adhesive, Masking, Paper
PPP-B-566	- Boxes, Folding, Paperboard
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-B-676	- Boxes, Setup

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Military

MIL-P-116	- Preservation-Packaging, Methods of
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Testing

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U.S. Military Property
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods

DRAWINGS

Rock Island Arsenal

C5506661	- Gun, Filler, Oil Screw, M3

PUBLICATIONS

Rock Island Arsenal

QEDL 5506661	- Quality Engineering Data List

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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National Bureau of Standards (NBS) Handbook

H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

* Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

* 3.1 First article. Unless otherwise specified (see 6.2.1), the supplier shall furnish one complete gun for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Design. The gun shall be in accordance with the design established by detail and assembly drawing C5506661. Parts subject to wear, breakage, or distortion shall be accessible for repair.

3.2.1 Lubrication. All working parts shall be provided with means for lubrication.

* 3.2.2 Threads. All threaded parts shall be in the inch system and shall comply with NBS H28 or applicable drawings.

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3.2.3 Interchangeability. All parts shall be manufactured to such standards as will permit replacement and adjustment without modification of parts.

3.3 Materials. Materials shall be as specified herein, and as specified on the applicable drawing and specifications. Materials not specifically designated shall be of good commercial quality. Contact between dissimilar metals, especially brass, copper, or steel in contact with aluminum or magnesium, shall be avoided wherever possible.

* 3.4 Construction. The gun shall be complete so it can be used for the operations specified herein. Parts and materials used in fabrication of the gun shall be new and free of defects. All surfaces of all parts shall be cleaned and free of extraneous materials. External surfaces shall be smooth and free of sharp edges. Functional surfaces shall be free from burrs and defects affecting proper performance. All screws, pins, and similar parts shall be installed to prevent loss of tightness. However, such parts subject to adjustment or removal shall not be permanently deformed. There shall be no processes used for reclaiming any part of the gun.

* 3.4.1 Protective finish. The final protective finish shall be as specified on drawing C5506661.

3.4.2 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching.

3.4.3 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where these operations are required in the original design.

3.5 Performance and product characteristics.

* 3.5.1 Performance. The gun shall be capable of being filled with hydraulic fluid (without air inclusion) by retraction of the screw when the nozzle is submerged in fluid conforming to MIL-H-6083. After being completely filled, the gun shall be capable of delivering this fluid from the nozzle at a pressure of 2,000 pound-force per square inch gage (psig). Leakage from the pressure chamber of the gun shall not exceed the following: 10 percent pressure or 1 percent volume for a period of 1 hour at 2,000 psig, while being operated within the temperature ranges specified in 3.5.2. The screw assembly shall operate its entire length of travel without binding, sticking, or chatter. The handle assembly shall function throughout its working length without binding and shall be retained in the screw assembly by stops.

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* 3.5.2 Operating temperature. The gun and all components shall conform to the performance requirements of 3.5.1 without damage while operating at a stabilized temperature through the range of -40°Fahrenheit (F) to +130°F.

* 3.6 Identification marking. The gun shall be marked for identification in accordance with MIL-STD-130 and unless otherwise specified (see 6.2.1), shall include the National Stock Number.

* 3.7 Technical data. Data shall be furnished as specified (see 6.2.2).

* 3.8 Workmanship. Workmanship shall be first-class to assure procurement of good quality equipment of a neat general appearance. Standards of workmanship shall be such that equipment specified herein will comply with the requirements in section 3. All components, subassemblies, parts, and the final assembly shall comply with tests in section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).

(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be applied to the preproduction model or initial production item noted in 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.3 and all tests under 4.4 Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

* 4.2.2 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production

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units offered for acceptance under the contract and shall consist of (a) through (d) as follows. Sampling will be conducted in accordance with MIL-STD-105 at the inspection level and AQL specified in (b) through (d).

(a) Product examination (see 4.3). Sampling not applicable. Each complete, assembled gun shall be examined. Failure to pass the examination shall be cause for rejection of that item.

(b) Component inspection (see 4.3.1). Inspection of components shall be in accordance with General Level II of MIL-STD-105. Acceptance to be based on the AQL value specified on applicable Supplementary Quality Assurance Provisions (SQAP) listed on Quality Engineering Data List (QEDL) 5506661 for acceptance inspection.

(c) Performance test (see 4.4.1). Sampling for test shall be in accordance with Special Inspection Level S-4. Failure of any item to pass the test shall be cause for rejection of the lot which the sample represents. NOTE: For the purpose of quality conformance inspection only, the test in 4.4.1 shall be conducted at a temperature range of +70°F ±10°F.

(d) Preparation for delivery inspection (see 4.5). Sampling for inspection shall be in accordance with Special Inspection Level S-4. Acceptance will be based on AQL of 2.5.

* 4.3 Product examination. Each gun (without fluid) shall be examined for completeness of components, proper assembly, and marking in accordance with drawing C5506661. The head retaining screw shall be checked for tightness. The screw assembly and handle assembly shall be examined to determine compliance with 3.5.1.

* 4.3.1 Component Inspection. Individual components of the gun shall be inspected according to technical requirements specified on the applicable SQAP listed on QEDL 5506661.

4.4 Test methods.

* 4.4.1 Performance test. Presoaking of the gun components with hydraulic fluid prior to testing is permissible. The gun shall be filled with hydraulic fluid (without air inclusion) by retraction of the screw when the nozzle is submerged in fluid conforming to MIL-H-6083. After being completely filled, the gun shall deliver the fluid from the nozzle counteracting a pressure of 2,000 psig. The screw assembly shall operate its entire length of travel without binding, sticking, or chatter. Leakage from the pressure chamber shall not exceed the following: 10 percent pressure or 1 percent volume for a period of 1 hour at temperature ranges specified in 4.4.2.1 and 4.4.2.2 to determine compliance with 3.5.1.

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* 4.4.2 Operating temperature test.

4.4.2.1 Low temperature operating test. The gun, after being in an ambient air temperature of -40°F to -45°F until stabilized and while at that temperature, shall be tested for performance in accordance with 4.4.1 to determine compliance with 3.5.2.

4.4.2.2 High temperature operating test. The gun, after being in an ambient air temperature of +130°F to +135°F until stabilized and while at that temperature, shall be tested for performance in accordance with 4.4.1 to determine compliance with 3.5.2.

4.5 Preparation for delivery inspection. Preservation-packaging, packing, and marking for shipment and storage shall be inspected to determine compliance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A-Maximum Military or Commercial, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The gun shall be cleaned in accordance with process C-5 of MIL-P-116 and dried in accordance with any applicable procedure specified therein.

5.1.1.2 Preservation application. Immediately after cleaning and drying, submerge the gun in preservative oil conforming to type P-7 of MIL-P-116, making certain that all surfaces, interior and exterior, are adequately coated. Remove gun from oil and allow to drain.

5.1.1.3 Unit packaging. The method of unit protection shall conform to method I criterion of MIL-P-116. To conserve cubage, remove the handle assembly from the basic gun and wrap handle with barrier material conforming to MIL-B-121, type II, grade A. Place the wrapped handle adjacent to the gun body and wrap the complete gun with similar type barrier material. Secure barrier material in place with tape conforming to UU-T-106. Place the gun in a snug-fitting box conforming to PPP-B-566 or PPP-B-676. The box closure shall be in accordance with the appendix to the applicable box specification.

5.1.1.4 Technical data. Technical data issued with the gun shall be packaged in accordance with method IC-1 criterion of MIL-P-116 and placed inside the unit container.

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5.1.1.5 Intermediate packaging. A quantity of unit packaged guns shall be placed in a close-fitting fiberboard box conforming to PPP-B-636, grade W6c. The box shall not exceed 40 pounds gross weight. The containers shall contain like quantities, as near as practicable, and be closed in accordance with the appendix to the box specification.

5.1.2 Commercial. The gun shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity. The package and quantity per package shall be that normally used by the supplier for retail distribution.

5.2 Packing. Packing shall be level A-Maximum Military, level B-Minimum Military, or Commercial, as specified (see 6.2.1).

5.2.1 Level A. The packaged guns shall be packed in a snug-fitting container conforming to PPP-B-601, overseas type or PPP-B-621, class 2. Closure and strapping requirements shall be in accordance with the appendix to the applicable container specification. The gross weight of the shipping container shall be governed by the limitations of the applicable container specification. When the container is not filled to its capacity, blocking shall be applied in accordance with MIL-STD-1186 to restrict movement of the contents.

5.2.2 Level B. The guns shall be packed as specified in 5.2.1, except the shipping container shall conform to PPP-B-640, class 2.

5.2.3 Commercial. The guns shall be packed to meet carrier acceptance and to assure safe delivery at destination, at lowest rates. Shipping containers shall comply with Uniform Freight Classification or National Motor Freight Classification Rules, or with rules and regulations of other carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 Military. When military levels of protection are specified, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 and any special markings required by the contract or purchase order.

5.3.2 Commercial. When commercial protection is specified, all unit and exterior packages and packs shall as a minimum be marked as follows:

(a) Noun nomenclature cited in the contract or purchase order.

(b) National Stock Number, or part number when NSN is not cited in the contract or purchase order.

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(c) Government contract or purchase order number.

(d) Quantity.

(e) Contractor's name.

(f) Additional markings as may be required by the contract and contractor's policy and procedures.

(g) Exterior containers shall also be marked with the appropriate address markings.

(h) Markings may be applied by any means which provide legibility.

6. NOTES

6.1 Intended use. This specification covers a screw primed, direct delivery gun to be used in connection with hydro-pneumatic recoil mechanisms and equilibrators.

* 6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

* 6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.

(b) First article, when required (see 3.1).

(c) Identification marking, if different (see 3.6).

(d) First article inspection, if different (see 4.2.1).

(e) Quality conformance inspection, if different (see 4.2.2).

(f) Selection of applicable levels of preservation-packaging, and packing (see 5.1 and 5.2).

* 6.2.2 Contract data requirements. Required technical data, such as operating instructions, parts lists, drawings, safety precautions, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract (see 3.7).

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6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army - WC

Review activity:
Army - SM

User activity:
Army - None

Preparing activity:
Army - WC

Project No. 4933-A025

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DOCUMENT IDENTIFIER AND TITLE

MIL-G-1349C(WC); Gun, Fluid, Direct Delivery

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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